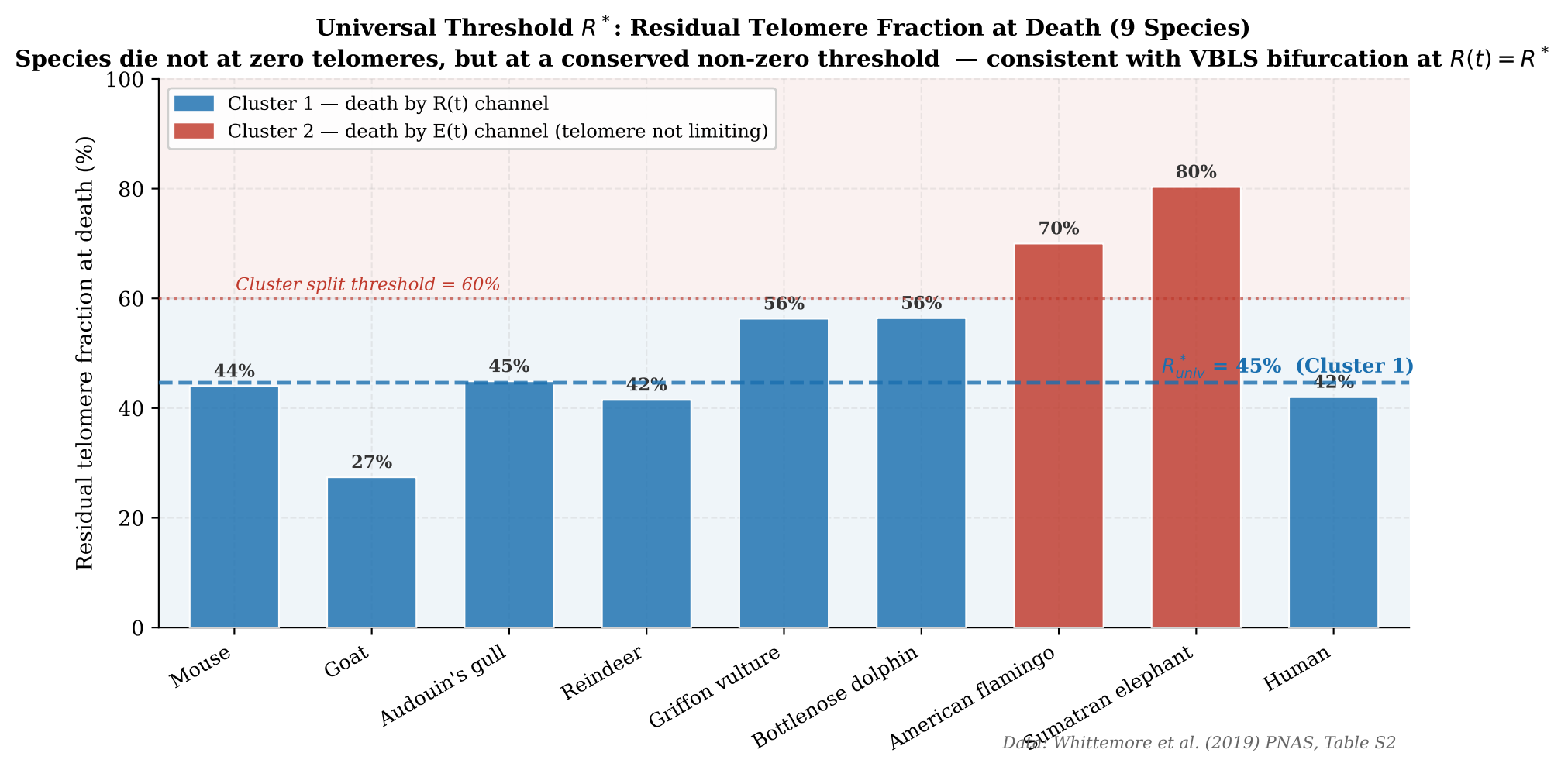
Universal Threshold R^*: Residual Telomere Fraction at Death (9 Species)
Species die not at zero telomeres, but at a conserved non-zero threshold — consistent with VBLS bifurcation at $R(t) = R^*$
Cluster 1 — death by R(t) channel
Cluster 2 — death by E(t) channel (telomere not limiting)
Residual telomere fraction at death (%)
Cluster split threshold = 60%
R^*_{univ} = 45% (Cluster 1)
44%
27%
45%
42%
56%
56%
70%
80%
42%
Mouse
Goat
Audouin's gull
Reindeer
Griffon vulture
Bottlenose dolphin
American flamingo
Sumatran elephant
Human
Data: Whittemore et al. (2019) PNAS, Table S2